Explicit dissociated sets from tensor lattices: an effective form of the disproof of Erdős's distinct subset sums conjecture

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Abstract

A finite set A of positive integers is *dissociated* if its $2^{|A|}$ subset sums are pairwise distinct. Erdős conjectured in 1931 that $\max A \geqslant c\,2^{|A|}$ for an absolute constant $c > 0$. Until 2026, every improvement of the trivial bound $\max A \leqslant 2^{|A|-1}$ came from an explicit construction: Conway and Guy (1968) reached $\max A < 0.23513 \cdot 2^{|A|}$, Lunnon (1988) $0.22096 \cdot 2^{|A|}$, and Bohman (1998) $0.22002 \cdot 2^{|A|}$, in each case for all sufficiently large $|A|$. In September 2026 the conjecture was disproved by an iterated lattice construction (GPT-6 Astra, expounded by Bloom): for every $\varepsilon > 0$ there are arbitrarily large n and dissociated A with $|A| = n$ and $\max A < \varepsilon 2^n$. That proof is non-effective, because it approximates a lattice of rank $d-1$ by a primitive one through a density argument.

We make the construction explicit. We describe the lattices in closed form inside $\mathbb{Z}[x_1, \ldots, x_s]/(x_1^b - 1, \ldots, x_s^b - 1)$, and we prove that adding suitable unit vectors to a basis of the scaled lattice produces an integer lattice whose normal vector is a tensor product $\tilde{u}_1 \otimes \cdots \otimes \tilde{u}_s$ of explicit vectors $\tilde{u}_k$ proportional to $x^{b-1} \operatorname{adj}(\overline{\varphi_k})$, where $\varphi_k = \theta_k(2+x)(1-x) + \rho_k$ are explicit "level polynomials" and adj is the adjugate in $\mathbb{Z}[x]/(x^b-1)$; the index of this lattice in its saturation is $\prod_k \operatorname{cont}(\operatorname{adj}\overline{\varphi_k})^{b^{s-k}}$, where cont denotes the content (the gcd of the coefficients). A level is primitive if and only if $\deg\gcd(\varphi_k, x^b - 1) \leqslant 1$ over every prime field, and the exceptional primes divide either $\gcd(\theta_k, \rho_k)$ or a fixed integer depending only on b. We deduce an unconditional effective theorem for one level, and, for every (b, s) satisfying a finite decidable coprimality condition (verified for eight pairs), an infinite arithmetic progression of good scalings and the bound $\limsup_{l\to\infty} f(l\,b^s) \leqslant (1 + 2^{-b})^{(b^s-1)/(b-1)}(2/3)^s$, where $f(n)$ is the infimum of $\max A/2^{n-1}$ over dissociated sets of size n. We give computer-certified explicit dissociated sets, for instance one with $|A| = 43\,923$ and $\max A < 0.15853 \cdot 2^{|A|}$ and one with $|A| = 3\,841\,966$ and $\max A < 0.10304 \cdot 2^{|A|}$; these are the first explicit sets below Bohman's constant. Since f is nonincreasing

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(double every element and adjoin an odd number), each of these bounds persists for every larger size, so that for instance $f(n) \leqslant 0.206072$ for all $n \geqslant 3\,841\,966$. Under the coprimality conjecture the method yields $f(n) \leqslant n^{-c/\log\log n}$ for all sufficiently large n. Finally we show that the exponent of decay is governed by the heights of admissible pairs of lattices.

Mathematics Subject Classifications: 11B75, 11H06, 11H31, 11Y16

1 Introduction

1.1 Background

Let $A \subset \mathbb{N}$ be finite. We call A *dissociated* (or say that A has *distinct subset sums*) if the map $S \mapsto \sum_{a \in S} a$ is injective on the subsets $S \subseteq A$. Erdős, who called this "perhaps my first serious problem", conjectured that every dissociated $A \subseteq \{1, \ldots, N\}$ with $|A| = n$ satisfies $N \gg 2^n$, and offered \$500 for a proof or disproof; it is Problem 1 of the collection [2]. Writing

$$f(n) := \inf\left\{\frac{\max A}{2^{n-1}} : A \subset \mathbb{N} \text{ dissociated}, \ |A| = n\right\},$$

the powers of two give $f(n) \leqslant 1$. Until 2026, every improvement of this trivial bound came from an explicit construction, and each such bound propagates from one size to all larger sizes because f is nonincreasing (Lemma 22 below). The Conway–Guy sequence [6, 10], shown to consist of dissociated sets by Bohman [3], gives $N < 0.23513 \cdot 2^n$ for $n \geqslant 40$; Lunnon's computer search [12] gives $N < 0.22096 \cdot 2^n$ for $n \geqslant 67$; and Bohman [4] obtained $f(n) \leqslant 0.44004$ for large n, i.e. $N \leqslant 0.22002 \cdot 2^n$, the best explicit construction prior to the present paper. In the other direction Erdős and Moser [9] proved $N \geqslant (\frac{1}{4} - o(1))2^n/\sqrt{n}$; the constant was improved several times (Elkies [8], Aliev [1]; see the table in [13] for the full history), the current record $\sqrt{2/\pi}$ being due to Elkies and Gleason (unpublished) and to Dubroff, Fox and Xu [7], who prove $N \geqslant \binom{n}{\lfloor n/2 \rfloor}$.

In September 2026 the conjecture was *disproved*: for every $\varepsilon > 0$ there are arbitrarily large n with $f(n) < \varepsilon$. The construction is due to the AI system GPT-6 Astra; we follow the exposition of Bloom [2], whose notation we adopt. The proof has two parts. The first is an elementary iterative construction of lattices $\Lambda_s \subset V_d = \{x \in \mathbb{R}^d : \sum_i x_i = 0\}$, $d = b^s$, which avoid the open unit cube and have small covolume. The second is the passage from Λ_s to an integer vector with dissociated coordinates: one needs a *primitive* lattice $L = u^\perp \cap \mathbb{Z}^d$ close to $Q\Lambda_s$, and its existence was derived from the equidistribution of primitive lattices in the space of lattices [11], which is non-effective. Bloom remarks that an effective version "could prove something like $f(n) \leqslant n^{-c/\log\log n}$ for infinitely many n". We note in passing that, combined with the classical monotonicity of f (Lemma 22), the disproof gives $f(n) \to 0$ along all n; what it does not provide is a single explicit set, an explicit size n, or an explicit rate.

1.2 Results

This paper carries out the second step explicitly. Throughout, $b \geqslant 5$ is odd with $3 \nmid b$, $s \geqslant 1$, $d = b^s$, $R_s = \mathbb{Z}[x_1, \dots, x_s]/(x_1^b - 1, \dots, x_s^b - 1) \cong \mathbb{Z}^d$, $g(x) = (2+x)(1-x) = 2 - x - x^2$, and

$$\Delta_{b,s} := \frac{(1 + 2^{-b})^{(b^s-1)/(b-1)}}{(3/2)^s} \qquad (1)$$

is the normalised covolume $\mathrm{covol}(\Lambda_s)/\sqrt{d}$ of the construction (Proposition 8).

Closed form of the lattices. We show (Proposition 8) that

$$\Lambda_s = \bigoplus_{k=1}^{s} 2^{-k}(2 + x_1) \cdots (2 + x_k)(1 - x_k)\, \mathbb{Z}[x_k, \dots, x_s], \qquad v_s = 2^{-s} \prod_{k=1}^{s} (2 + x_k),$$

where $\mathbb{Z}[x_k, \dots, x_s] \subset R_s$ is the span of the monomials not involving $x_1, \dots, x_{k-1}$.

An explicit primitivisation. For $Q \in 2^s\mathbb{Z}$ we add to each basis vector $Q\, 2^{-k}(2 + x_1) \cdots (2 + x_k)(1 - x_k)x_k^i m$ of $Q\Lambda_s$ ($0 \leqslant i \leqslant b - 2$, m a monomial in $x_{k+1}, \dots, x_s$) the monomial $x_1^{b-1} \cdots x_{k-1}^{b-1} x_k^i m$. This defines a lattice $L_s(Q) \subset \mathbb{Z}^d$ of rank $d - 1$. Define *level polynomials* $\varphi_k = \theta_k g + \rho_k \in \mathbb{Z}[x]/(x^b - 1)$ recursively by $\theta_1 = Q/2$, $\rho_1 = 1$, $\psi_k = \mathrm{adj}(\overline{\varphi_k})$ (the adjugate element, $\psi_k \overline{\varphi_k} = \mathrm{N}(\varphi_k)$), $\gamma_k = \mathrm{cont}(\psi_k)$, $\tilde{u}_k = \pm x^{b-1}\psi_k/\gamma_k$, $a_k = \langle \tilde{u}_k, x^{b-1} \rangle$, $c_k = \langle \tilde{u}_k, 2 + x \rangle$, $\theta_{k+1} = 2^{-(k+1)}Q\, c_1 \cdots c_k$, $\rho_{k+1} = a_1 \cdots a_k$ (Definition 13). The constant K_s below is explicit (Proposition 11).

Theorem 1 (normal vector and index)**.** *Let $Q \geqslant 8K_s$. Then $L_s(Q)$ has rank $d - 1$, its orthogonal complement is spanned by the primitive integer vector $u = \tilde{u}_1 \otimes \cdots \otimes \tilde{u}_s$, i.e. $u(x) = \prod_k \tilde{u}_k(x_k)$, all coordinates of u are positive, and*

$$[\,\mathrm{sat}\, L_s(Q) : L_s(Q)\,] = \prod_{k=1}^{s} \gamma_k^{b^{s-k}}.$$

In particular $L_s(Q)$ is primitive if and only if $\gamma_k = \mathrm{cont}\,\mathrm{adj}(\overline{\varphi_k}) = 1$ for every k.

Theorem 2 (level criterion and exceptional primes)**.** *For $\varphi = \theta g + \rho \in \mathbb{Z}[x]/(x^b - 1)$ with* $\mathrm{N}(\varphi) \neq 0$, *one has* $\mathrm{cont}\,\mathrm{adj}(\bar{\varphi}) = 1$ *if and only if* $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) \leqslant 1$ *for every prime p. Let P_b be the set of odd primes dividing $b(2^b + 1)n_b$, where $n_b = \prod_{0<i<j<b} \mathrm{Res}_z\big(\frac{z^b-1}{z-1},\ 1 + z^i + z^j\big) \neq 0$. If p is an odd prime with $p \notin P_b$ and $p \nmid \gcd(\theta, \rho)$, then* $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) \leqslant 1$.

Theorem 3 (first level, unconditional)**.** *Let $q \geqslant 4K_1$ be an integer with* $\mathrm{rad}(P_b) \mid q$. *Then $L_1(2q)$ is primitive; the coordinates of $\tilde{u}_1$ are positive integers forming a $(2q - K_1)$-dissociated set, with* $\max \tilde{u}_1 \leqslant (2q)^{b-1}\Delta_{b,1}(1 + 5K_1/q)$. *Consequently, for every l with $2^l \leqslant 2q - K_1$ the set $\{2^j \tilde{u}_{1,i} : 0 \leqslant j < l,\ 0 \leqslant i < b\}$ is dissociated of size lb, and* $f(lb) \leqslant \max \tilde{u}_1 / 2^{l(b-1)}$.

Explicit dissociated sets. Theorem 1, a perturbation lemma (Lemma 9) and a size estimate (Lemma 10) together give the following.

Theorem 4. *Let $2^s \mid Q$, $Q \geqslant 8K_s$, and suppose $\gamma_k = 1$ for $k = 1, \ldots, s$ (a finite computation). Then the $d = b^s$ coordinates of u are positive integers forming a $(Q - K_s)$-dissociated set, and*

$$\max u = \prod_{k=1}^{s} \max \tilde{u}_k \;\leqslant\; Q^{d-1} \Delta_{b,s} \, (1 + 10K_s/Q).$$

For every l with $2^l \leqslant Q - K_s$ the set $A = \{2^j u_i\}$ is dissociated, $|A| = ld$, and $f(ld) \leqslant \max u / 2^{l(d-1)}$.

The certificate $\gamma_1 = \cdots = \gamma_s = 1$ costs s adjugate computations of $b \times b$ integer matrices, so it is cheap even when $d = b^s$ is in the tens of thousands. Table 1 lists the outcomes; for example

$$\begin{aligned}
(b, s) = (11, 3): &\quad n = 43\,923, &\quad \max A < 0.317045 \cdot 2^{n-1} = 0.158523 \cdot 2^n, \\
(b, s) = (13, 4): &\quad n = 1\,199\,562, &\quad \max A < 0.265069 \cdot 2^{n-1} = 0.132535 \cdot 2^n, \\
(b, s) = (17, 4): &\quad n = 3\,841\,966, &\quad \max A < 0.206072 \cdot 2^{n-1} = 0.103036 \cdot 2^n.
\end{aligned}$$

To our knowledge these are the first explicit dissociated sets beating Bohman's constant 0.22002. By Lemma 22 each bound persists for all larger sizes: there are explicit dissociated sets with $\max A < 0.158523 \cdot 2^n$ for every $n \geqslant 43\,923$ and with $\max A < 0.103036 \cdot 2^n$ for every $n \geqslant 3\,841\,966$ (Corollary 23).

Infinitely many good scalings. Write $Q = 2^{s+1}t$. Before contents are normalised, the level data $\hat{a}_k, \hat{c}_k$ are polynomials in t with integer coefficients (Section 5.1). Let $H(b, s)$ be the statement that $\hat{a}_k$ and $\hat{c}_k$ are coprime in $\mathbb{Q}[t]$ for $1 \leqslant k \leqslant s - 1$.

Theorem 5. *Assume $H(b, s)$. There is an integer $M \geqslant 1$ such that $L_s(2^{s+1}t)$ is primitive for every $t \equiv 0 \pmod{M}$ with $2^{s+1}t \geqslant 8K_s$. Consequently $\limsup_{l \to \infty} f(l\, b^s) \leqslant \Delta_{b,s}$.*

$H(b, s)$ is a finite, decidable statement, and we have verified it for

$$(b, s) \in \{(5, 2), (7, 2), (11, 2), (13, 2), (5, 3), (7, 3), (11, 3), (13, 3)\}$$

(Section 6); hence $\limsup_l f(1331\, l) \leqslant \Delta_{11,3} < 0.3162$ and $\limsup_l f(2197\, l) \leqslant \Delta_{13,3} < 0.303$ unconditionally (Corollary 27). If $H(b, s)$ holds for all (b, s) (Conjecture 28), the construction gives $f(n) \leqslant n^{-c/\log\log n}$ for infinitely many n with an explicit $c > 0$ (Theorem 29, whose proof we only sketch), and then, by monotonicity, for all sufficiently large n with a smaller explicit constant (Corollary 30).

Two structural remarks. The sizes $n = ld$ reached by the construction itself form a sparse set, and in Section 7 we explain why the natural padding operations *inside* the construction lose a factor exponential in the number of added elements; the gaps between consecutive sizes are filled by the classical doubling step (Lemma 22) rather than by the lattice construction. We also show that the exponent of decay is governed by the *height* of admissible pairs: an admissible pair in dimension b with height $h \geqslant b^{\alpha}$ and covolume $1 + O(b^{-Cs})$ would give $f(n) \leqslant n^{-\alpha + o(1)}$, while heights are $O(\sqrt{b})$ (Proposition 32).

Organisation. Section 2 recasts the lattices of [2] in the polynomial model R_s, in which a cyclic shift of coordinates is multiplication by a variable, and proves the closed form. Section 3 contains the two perturbation lemmas and the constant K_s. Section 4 defines $L_s(Q)$ and proves Theorems 1, 2 and 3. Section 5 deduces Theorems 4 and 5, the monotonicity lemma with its corollaries, and the conditional rate. Section 6 describes the computations behind Table 1, and Section 7 discusses the sizes reached by the construction, the height problem and Siegel's lemma.

1.3 Notation and conventions

For a lattice L of rank r in $\mathbb{R}^d$, $\operatorname{covol}(L)$ is the r-dimensional volume of a fundamental domain. A sublattice $L \subseteq \mathbb{Z}^d$ is *primitive* (or saturated) if $L = (L \otimes \mathbb{R}) \cap \mathbb{Z}^d$; we write $\operatorname{sat} L$ for $(L \otimes \mathbb{R}) \cap \mathbb{Z}^d$. If $L \subset \mathbb{Z}^d$ has rank $d-1$ with basis matrix $B \in \mathbb{Z}^{d\times(d-1)}$, the vector w of signed maximal minors of B spans $L^\perp$ and $\gcd(w) = [\operatorname{sat} L : L]$; a primitive lattice of rank $d-1$ is $R(u) := u^\perp \cap \mathbb{Z}^d$ for a primitive $u \in \mathbb{Z}^d$, and $\operatorname{covol}(R(u)) = |u|_2$. For $u \in \mathbb{Z}^d$ and real $Q > 0$, the coordinates of u are *Q-dissociated* if the only $c \in \mathbb{Z}^d$ with $|c_i| < Q$ for all i and $\sum_i c_i u_i = 0$ is $c = 0$; equivalently $R(u) \cap (-Q, Q)^d = \{0\}$. The *lift* of a 2^l-dissociated vector u with positive coordinates is $A = \{2^j u_i : 0 \leqslant j < l,\ 1 \leqslant i \leqslant d\}$; two subset sums of A are $\sum_i c_i u_i$ and $\sum_i c_i' u_i$ with $0 \leqslant c_i, c_i' < 2^l$, so A is dissociated, $|A| = ld$, $\max A = 2^{l-1} \max u$, and

$$f(ld) \leqslant \frac{2^{l-1} \max u}{2^{ld-1}} = \frac{\max u}{2^{l(d-1)}}. \tag{2}$$

$R_b = \mathbb{Z}[x]/(x^b - 1)$ is identified with $\mathbb{Z}^b$ via $x^i \leftrightarrow e_i$; the standard inner product becomes $\langle f, g\rangle = [x^0](f\bar{g})$, where $\bar{g}(x) = g(x^{-1})$ and $[x^i]h$ denotes the coefficient of x^i in h; in particular $\langle f, x^i\rangle = [x^i]f$. For $a \in R_b$ let m_a be the $b \times b$ matrix of multiplication by a (a circulant), $\mathrm{N}(a) = \det m_a = \prod_{\zeta^b=1} a(\zeta)$, and $\operatorname{adj}(a) \in R_b$ the element with $m_{\operatorname{adj}(a)} = \operatorname{adj}(m_a)$, so that $a \operatorname{adj}(a) = \mathrm{N}(a)$; since $\operatorname{adj}(m_a)$ commutes with every m_c, it is indeed a multiplication matrix. Note $m_a^T = m_{\bar{a}}$. The content $\operatorname{cont}(\psi)$ of $\psi \in R_b$ is the gcd of its coefficients. Finally $\mathbf{1}_d = (1, \dots, 1) \in \mathbb{R}^d$, $V_d = \mathbf{1}_d^\perp$, and μ_b is the group of complex b-th roots of unity.

2 The lattices

2.1 Admissible pairs

Following [2], a pair (Λ, v) with $\Lambda \subset V_d$ a lattice of rank $d-1$ and $v \in \mathbb{R}^d \setminus V_d$ is *admissible* if

(A1) $\Lambda \cap (-1, 1)^d = \{0\}$;

(A2) $tv \in \Lambda + (-1, 1)^d$ with $t \in \mathbb{R}$ implies $|t| < 1$.

Its *height* is $h = \langle v, \mathbf{1}_d\rangle$, and $\Gamma = \Lambda \oplus \mathbb{Z}v$ satisfies $\operatorname{covol}(\Gamma) = \operatorname{covol}(\Lambda)\, h/\sqrt{d}$. We write $\Delta = \operatorname{covol}(\Lambda)/\sqrt{d} = \operatorname{covol}(\Gamma)/h$. Conditions (A1)–(A2) imply $\Gamma \cap (-1,1)^d = \{0\}$, hence $\operatorname{covol}(\Gamma) \geqslant 1$ by Minkowski's theorem, so $\Delta \geqslant 1/h$.

The base pair. Let $b \geqslant 3$ be odd. Put

$$\Lambda_1 = \tfrac{1}{2} g\, R_b = \tfrac{1}{2}(2+x)\,\big((1-x)R_b\big), \qquad v_1 = \tfrac{1}{2}(2+x).$$

Since $(1-x)R_b = \mathbb{Z}^b \cap V_b$ and multiplication by $\frac{1}{2}(2+x)$ is the map $M = I + \frac{1}{2}P$ (P the cyclic shift), this is exactly the base pair of [2]: $h_1 = 3/2$, $\operatorname{covol}(\Gamma_1) = \det M = 1 + 2^{-b}$, $\Delta_{b,1} = \frac{2}{3}(1+2^{-b})$, and (Λ_1, v_1) is admissible ([2]; the proof uses only that b is odd).

2.2 Tensor products of admissible pairs

Proposition 6. *Let (Λ, v) be admissible in $\mathbb{R}^b$ and (Λ', v') admissible in $\mathbb{R}^{d'}$. In $\mathbb{R}^b \otimes \mathbb{R}^{d'} = \mathbb{R}^{bd'}$ put*

$$\Lambda'' = \Lambda \otimes \mathbb{Z}^{d'} \;\oplus\; v \otimes \Lambda', \qquad v'' = v \otimes v'.$$

Then (Λ'', v'') is admissible, of height hh', and $\operatorname{covol}(\Gamma'') = \operatorname{covol}(\Gamma)^{d'} \operatorname{covol}(\Gamma')$.

Proof. View elements of $\mathbb{R}^b \otimes \mathbb{R}^{d'}$ as $b \times d'$ matrices. An element of Λ'' has the form $X = \sum_i \beta_i z_i^T + v\lambda^T$ with $\beta_i \in \Lambda$, $z_i \in \mathbb{Z}^{d'}$, $\lambda \in \Lambda'$, so its m-th column is $\sum_i \beta_i z_{i,m} + \lambda_m v \in \Lambda + \lambda_m v$. Coordinate sums vanish, and $\Lambda \otimes \mathbb{R}^{d'}$, $v \otimes \mathbb{R}^{d'}$ intersect trivially, so Λ'' is a lattice of rank $(b-1)d' + (d'-1) = bd' - 1$ in $V_{bd'}$. If $X \in (-1,1)^{bd'}$, each column lies in $(\Lambda + \lambda_m v) \cap (-1,1)^b$, so $|\lambda_m| < 1$ by (A2) for (Λ, v); thus $\lambda \in \Lambda' \cap (-1,1)^{d'} = \{0\}$, and then every column lies in $\Lambda \cap (-1,1)^b = \{0\}$. If $X \in \Lambda'' + tv''$ has all entries in $(-1,1)1$, the m-th column lies in $\Lambda + (\lambda_m + tv'_m)v$, so $|\lambda_m + tv'_m| < 1$ for all m, i.e. $\lambda + tv' \in (\Lambda' + tv') \cap (-1,1)^{d'}$, and (A2) for (Λ', v') gives $|t| < 1$. The height is $\langle v, \mathbf{1}_b\rangle\langle v', \mathbf{1}_{d'}\rangle = hh'$. Finally $\Gamma'' = \Lambda \otimes \mathbb{Z}^{d'} \oplus v \otimes \Gamma'$; the first summand spans $V_b \otimes \mathbb{R}^{d'}$ and has covolume $\operatorname{covol}(\Lambda)^{d'}$, while the projection of $v \otimes \Gamma'$ onto $(V_b \otimes \mathbb{R}^{d'})^\perp = \mathbf{1}_b \otimes \mathbb{R}^{d'}$ is $\frac{h}{\sqrt{b}}\, \hat{n} \otimes \Gamma'$ with $\hat{n} = \mathbf{1}_b/\sqrt{b}$, of covolume $(h/\sqrt{b})^{d'} \operatorname{covol}(\Gamma')$. Hence $\operatorname{covol}(\Gamma'') = (\operatorname{covol}(\Lambda) h/\sqrt{b})^{d'} \operatorname{covol}(\Gamma') = \operatorname{covol}(\Gamma)^{d'} \operatorname{covol}(\Gamma')$. $\square$

The iteration of [2] is the same operation with the roles of the two factors exchanged; both produce admissible pairs with identical invariants. We use the order of Proposition 6 because it gives a clean closed form.

Definition 7. (Λ_1, v_1) is the base pair in the variable x_1; for $s \geqslant 2$, $(\Lambda_s, v_s) := (\Lambda_1, v_1) \otimes (\Lambda_{s-1}, v_{s-1})$, where Λ_{s-1} lives in the variables $x_2, \dots, x_s$.

Proposition 8 (closed form)**.** *Let $\mathbb{Z}[x_k, \dots, x_s] \subset R_s$ be the span of the monomials in $x_k, \dots, x_s$. Then*

$$\Lambda_s = \bigoplus_{k=1}^{s} \Lambda_s^{(k)}, \qquad \Lambda_s^{(k)} = 2^{-k}(2+x_1)\cdots(2+x_k)(1-x_k)\,\mathbb{Z}[x_k, \dots, x_s],$$

$v_s = 2^{-s}\prod_{k=1}^{s}(2+x_k)$, $h_s = (3/2)^s$, $\operatorname{covol}(\Gamma_s) = (1+2^{-b})^{(b^s-1)/(b-1)}$ *and* $\operatorname{covol}(\Lambda_s)/\sqrt{b^s} = \Delta_{b,s}$ *as in (1). A basis of $\Lambda_s^{(k)}$ consists of the vectors $2^{-k}(2+x_1)\cdots(2+x_k)(1-x_k)\,x_k^i\, m$, $0 \leqslant i \leqslant b-2$, m a monomial in $x_{k+1}, \dots, x_s$; the denominators of Λ_s divide 2^s.*

Proof. Induction on s; $s=1$ is the definition of Λ_1, since $(1-x)R_b$ has basis $(1-x)x^i$, $i \leqslant b-2$. For $s \geqslant 2$, $\Lambda_1 \otimes \mathbb{Z}[x_2,\dots,x_s] = \frac{1}{2}(2+x_1)(1-x_1)R_s = \Lambda_s^{(1)}$ and $v_1 \otimes \Lambda_{s-1} = \frac{1}{2}(2+x_1)\bigoplus_{k\geqslant 2} 2^{-(k-1)}(2+x_2)\cdots(2+x_k)(1-x_k)\mathbb{Z}[x_k,\dots,x_s] = \bigoplus_{k\geqslant 2}\Lambda_s^{(k)}$ by the induction hypothesis. The invariants follow from Proposition 6: $h_s = \frac{3}{2}h_{s-1}$ and $\operatorname{covol}(\Gamma_s) = \operatorname{covol}(\Gamma_1)^{b^{s-1}}\operatorname{covol}(\Gamma_{s-1})$, which gives the exponent $1+b+\cdots+b^{s-1}$. □

3 Perturbation lemmas

Let (Λ, v) be admissible in $\mathbb{R}^d$, let $b_1,\dots,b_{d-1}$ be a fixed basis of Λ, and let $b_1^*,\dots,b_{d-1}^* \in V_d$ be the dual basis, so that $\langle b_i, b_j^*\rangle = \delta_{ij}$. Put

$$K := \sum_{j=1}^{d-1} |b_j^*|_1. \tag{3}$$

For $\lambda = \sum_j m_j b_j \in \Lambda$ we have $m_j = \langle \lambda, b_j^*\rangle$, hence

$$\sum_j |m_j| \leqslant K\,|\lambda|_\infty. \tag{4}$$

Let $Q>0$ be real, let $r_1,\dots,r_{d-1} \in \{1,\dots,d\}$ be *distinct* rows, and

$$L := \operatorname{span}_{\mathbb{Z}}\{\ell_j := Qb_j + e_{r_j} \ : \ 1 \leqslant j \leqslant d-1\} \subset \mathbb{R}^d.$$

Lemma 9 (cube avoidance)**.** *If $Q > K$ then the ℓ_j are linearly independent and $L \cap (-Q-K, Q-K)^d = \{0\}$. In particular, if $L \subseteq \mathbb{Z}^d$ is primitive with normal vector u, the coordinates of u are $(Q-K)$-dissociated.*

Proof. Let $x = \sum_j m_j \ell_j = Q\lambda + \sum_j m_j e_{r_j}$ with $\lambda = \sum_j m_j b_j$. If $\lambda = 0$ then all $m_j = 0$ and $x=0$. Otherwise $|\lambda|_\infty \geqslant 1$ by (A1), and by (4) $|x|_\infty \geqslant Q|\lambda|_\infty - \sum_j |m_j| \geqslant (Q-K)|\lambda|_\infty \geqslant Q-K > 0$. □

Lemma 10 (shape of the normal vector)**.** (i) *Let $Q > K$ and let $u \in \mathbb{R}^d\backslash\{0\}$ be orthogonal to L. Then $\langle u, \mathbf{1}_d\rangle \neq 0$ and, after replacing u by $-u$ if necessary, $u = \alpha(\mathbf{1}_d + \eta)$ with $\alpha > 0$, $\eta \in V_d$ and $|\eta|_\infty \leqslant K/(Q-K)$. In particular, if $Q > 2K$ all coordinates of u have the same sign.*

(ii) *If moreover $Q > 2K$, $Qb_j \in \mathbb{Z}^d$ for all j, and $u \in \mathbb{Z}^d$ is primitive, then*

$$\alpha \leqslant Q^{d-1}\Delta\, e^{K/Q}\Big(1 + \tfrac{K^2}{(Q-K)^2}\Big)^{1/2},$$
$$\max u \leqslant Q^{d-1}\Delta\, e^{K/Q}\Big(1 + \tfrac{K^2}{(Q-K)^2}\Big)^{1/2}\Big(1 + \tfrac{K}{Q-K}\Big) \leqslant Q^{d-1}\Delta\Big(1 + \frac{10K}{Q}\Big).$$

Proof. (i) Write $u = \alpha\mathbf{1}_d + \eta'$ with $\eta' \in V_d$. From $\langle u, \ell_j\rangle = 0$ and $\langle \mathbf{1}_d, b_j\rangle = 0$ we get $Q\langle \eta', b_j\rangle + \alpha + \eta'_{r_j} = 0$ for all j. If $\alpha = 0$ then $\eta' = \sum_j \langle \eta', b_j\rangle b_j^* = -Q^{-1}\sum_j \eta'_{r_j} b_j^*$, so

$|\eta'|_\infty \leqslant (K/Q)|\eta'|_\infty$ and $u = 0$, a contradiction. So $\alpha \neq 0$; take $\alpha > 0$ and $\eta = \eta'/\alpha$, so that $\langle \eta, b_j\rangle = -(1+\eta_{r_j})/Q$ and $\eta = -Q^{-1}\sum_j (1+\eta_{r_j})b_j^*$, whence $|\eta|_\infty \leqslant Q^{-1}(1+|\eta|_\infty)K$, i.e. $|\eta|_\infty \leqslant K/(Q-K)$, which is < 1 when $Q > 2K$.

(ii) Since $L \subseteq R(u)$, $\operatorname{covol}(L) \geqslant \operatorname{covol}(R(u)) = |u|_2 = \alpha|\mathbf{1}_d + \eta|_2 \geqslant \alpha\sqrt{d}$. The orthogonal projection $\pi\colon u^\perp \to V_d$ is bijective, as $\langle u, \mathbf{1}_d\rangle \neq 0$, and it multiplies covolumes by $\cos\vartheta = |\langle u, \mathbf{1}_d\rangle|/(|u|_2\sqrt{d}) = \sqrt{d}/|\mathbf{1}_d + \eta|_2 \geqslant (1+|\eta|_\infty^2)^{-1/2}$. Now $\pi(\ell_j) = Qb_j + \pi(e_{r_j})$ and $\pi(e_{r_j}) = \sum_i \langle e_{r_j}, b_i^*\rangle b_i = \sum_i (b_i^*)_{r_j} b_i$, so $\pi(L)$ has basis matrix $B(QI + A)$ with $A_{ij} = (b_i^*)_{r_j}$, where B is the basis matrix of Λ. Hence

$$\begin{aligned}\operatorname{covol}(\pi(L)) &= Q^{d-1}\operatorname{covol}(\Lambda)\,|\det(I + A/Q)|\\ &\leqslant Q^{d-1}\sqrt{d}\,\Delta\prod_i\Big(1 + \frac{|b_i^*|_1}{Q}\Big) \leqslant Q^{d-1}\sqrt{d}\,\Delta\, e^{K/Q}.\end{aligned}$$

Here the first inequality is Hadamard's, using $\sum_j |A_{ij}| \leqslant |b_i^*|_1$ (the rows r_j are distinct), and the second follows from (3). Therefore $\alpha\sqrt{d} \leqslant \operatorname{covol}(L) = \operatorname{covol}(\pi(L))/\cos\vartheta \leqslant Q^{d-1}\sqrt{d}\,\Delta\, e^{K/Q}(1+|\eta|_\infty^2)^{1/2}$. The last inequality of the lemma is elementary for $x = K/Q \leqslant \frac{1}{2}$: $e^x \leqslant 1 + 1.3x$, $(1 + x^2/(1-x)^2)^{1/2} \leqslant 1 + x$, $1 + x/(1-x) \leqslant 1 + 2x$, and $(1+1.3x)(1+x)(1+2x) \leqslant 1 + 10x$. ☐

Proposition 11 (the constant $\boldsymbol{K_s}$)**.** *Let $K_1 = \sum_{i\leqslant b-2} |b_i^*|_1$ for the base basis $b_i = \frac{1}{2}gx^i$ of Λ_1, and let $c_i = \langle b_i^*, v_1\rangle/h_1$. For the basis of Λ_s in Proposition 8, which is $\{b_i\otimes m\}\cup\{v_1\otimes b_j'\}$ with b_j' the basis of Λ_{s-1} and $d' = b^{s-1}$, a dual basis is*

$$(b_i\otimes m)^* = b_i^*\otimes m - c_i\,\mathbf{1}_b\otimes\big(m - \tfrac{1}{d'}\mathbf{1}_{d'}\big), \qquad (v_1\otimes b_j')^* = \tfrac{1}{h_1}\mathbf{1}_b\otimes(b_j')^*.$$

Consequently the constant (3) *for this basis of Λ_s equals*

$$K_s = d'\sum_i\Big(\big|b_i^* - c_i\big(1-\tfrac{1}{d'}\big)\mathbf{1}_b\big|_1 + |c_i|\,b\,\tfrac{d'-1}{d'}\Big) + \tfrac{b}{h_1}K_{s-1} \;\leqslant\; d'\big(1+\tfrac{4b}{3}\big)K_1 + \tfrac{2b}{3}K_{s-1}.$$

Proof. Direct verification of $\langle\cdot,\cdot\rangle = \delta$, using $\langle\mathbf{1}_b, b_i\rangle = 0$, $\langle\mathbf{1}_b, v_1\rangle = h_1$, $\langle m - \frac{1}{d'}\mathbf{1}_{d'}, b_j'\rangle = (b_j')_m$, and that all vectors lie in $V_{bd'}$. The ℓ^1 norms are read off blockwise; the crude bound uses $|c_i| \leqslant |b_i^*|_1|v_1|_\infty/h_1 = \frac{2}{3}|b_i^*|_1$. ☐

Numerically $K_1(5) = 424/55 \approx 7.71$, $K_1(7) \approx 14.19$, $K_1(11) \approx 32.20$, $K_1(13) \approx 44.03$, and, for instance, $K_2(5) \approx 98.25$, $K_3(11) \approx 1.78\cdot 10^4$, $K_4(13) \approx 5.52\cdot 10^5$. We checked the recursion against a direct computation of the dual basis for $(b,s) \in \{(5,2),(7,2),(5,3)\}$.

4 The tensor perturbation

Fix $b \geqslant 5$ odd with $3 \nmid b$, $s \geqslant 1$, $d = b^s$, and R_s, $x_1,\dots,x_s$ as in Proposition 8.

Definition 12 (the lattice $\boldsymbol{L_s(Q,\rho)}$)**.** Let $Q \in 2^s\mathbb{Z}_{>0}$ and $\rho \in \mathbb{Z}_{>0}$. For $1 \leqslant k \leqslant s$ put

$$\Phi_k = \Phi_k^{(Q,\rho)} := \frac{Q}{2^k}(1-x_k)\prod_{j\leqslant k}(2+x_j) \;+\; \rho\prod_{j<k}x_j^{b-1} \;\in R_s,$$

and let $L_s(Q,\rho)$ be the $\mathbb{Z}$-span of the $d-1$ vectors $\ell_{k,i,m} := x_k^i\, m\, \Phi_k$, where $1 \leqslant k \leqslant s$, $0 \leqslant i \leqslant b-2$, and m runs over the monomials in $x_{k+1}, \dots, x_s$. We write $L_s(Q) = L_s(Q,1)$.

By Proposition 8, $\ell_{k,i,m} = Q\,\beta_{k,i,m} + \rho\,\mu_{k,i,m}$, where the $\beta_{k,i,m}$ form the basis of Λ_s and $\mu_{k,i,m} = x_1^{b-1} \cdots x_{k-1}^{b-1} x_k^i m$ are *distinct* monomials (the variable x_k is the first one with exponent $\leqslant b-2$), none of them equal to $\prod_j x_j^{b-1}$. Thus $L_s(Q)$ is exactly the lattice of Section 3 for Λ_s with the tensor basis and distinct rows, and $L_s(Q,\rho) = \rho \cdot \operatorname{span}\{(Q/\rho)\beta_{k,i,m} + \mu_{k,i,m}\}$ is the same construction with the real scale Q/ρ.

Definition 13 (level data)**.** Set $\theta_1 = Q/2$ and $\rho_1 = \rho$. For $k \geqslant 1$, provided $\theta_k, \rho_k > 0$, put

$$
\begin{gathered}
\varphi_k = \theta_k g + \rho_k \in R_b, \qquad \mathrm{N}_k = \mathrm{N}(\varphi_k), \qquad \psi_k = \operatorname{adj}(\overline{\varphi_k}) \quad (\text{so } \psi_k \overline{\varphi_k} = \mathrm{N}_k), \\
\gamma_k = \operatorname{cont}(\psi_k), \qquad \tilde{u}_k = \varepsilon_k\, x^{b-1} \psi_k / \gamma_k \quad (\varepsilon_k \in \{\pm 1\} \text{ with } \langle \tilde{u}_k, \mathbf{1}_b \rangle > 0), \\
a_k = \langle \tilde{u}_k, x^{b-1} \rangle = \varepsilon_k \psi_{k,0}/\gamma_k, \qquad c_k = \langle \tilde{u}_k, 2 + x \rangle = 2\tilde{u}_{k,0} + \tilde{u}_{k,1}, \\
\theta_{k+1} = \frac{Q}{2^{k+1}}\, c_1 \cdots c_k, \qquad \rho_{k+1} = \rho\, a_1 \cdots a_k,
\end{gathered}
$$

where $\tilde{u}_{k,i}$ and $\psi_{k,i}$ denote the coefficients of x^i; thus $\tilde{u}_{k,i} = \varepsilon_k \psi_{k,i+1}/\gamma_k$ with indices mod b.

Lemma 14 (non-degeneracy)**.** *If $Q \geqslant 8K_s\rho$, then for every $k \leqslant s$: $\theta_k, \rho_k > 0$, $\mathrm{N}_k \neq 0$, and $\tilde{u}_k$ has all coordinates positive (so $a_k, c_k > 0$). Moreover $2\theta_k/\rho_k \geqslant Q/\rho$.*

Proof. For $\theta, \rho' > 0$ the element $\theta g + \rho'$ is not a zero divisor of $R_b \otimes \mathbb{Q}$: a root $\zeta \in \mu_b$ would satisfy $g(\zeta) = -\rho'/\theta \in \mathbb{Q}_{<0}$, but $g(\zeta) = 2 - \zeta - \zeta^2$ is rational only for $\zeta \in \{1, -1, \omega, \bar{\omega}\}$ (ω a primitive cube root of unity), and $\mu_b \cap \{1, -1, \omega, \bar{\omega}\} = \{1\}$ because b is odd and $3 \nmid b$, while $g(1) = 0$. Now induct on k. Given $\theta_k, \rho_k > 0$ we have $\mathrm{N}_k \neq 0$ and $\psi_k \neq 0$. The lattice $L^{(k)} := \operatorname{span}\{x^i \varphi_k : i \leqslant b-2\} = \rho_k \operatorname{span}\{(2\theta_k/\rho_k) b_i + e_i\}$ is of the type of Lemma 10 with $K = K_1$ and real scale $Q'_k := 2\theta_k/\rho_k$, and its orthogonal complement is spanned by $x^{b-1}\psi_k$ (Proposition 15 below). If $Q'_k \geqslant 8K_1$, Lemma 10(i) shows that $\tilde{u}_k$ has coordinates of one sign, all in $[\alpha(1 - \frac{1}{7}), \alpha(1 + \frac{1}{7})]$, whence $c_k/a_k \geqslant 3 \cdot \frac{6/7}{8/7} > 2$. Since $K_s \geqslant (2b/3)^{s-1} K_1 \geqslant K_1$ by Proposition 11, we have $Q'_1 = Q/\rho \geqslant 8K_1$, and inductively $2\theta_{k+1}/\rho_{k+1} = (Q/2^k \rho) \prod_{j \leqslant k} (c_j/a_j) \geqslant Q/\rho \geqslant 8K_1$. $\square$

Proposition 15 (one level)**.** *Let $\varphi \in R_b$ with $\mathrm{N}(\varphi) \neq 0$ and $L = \operatorname{span}\{x^i \varphi : 0 \leqslant i \leqslant b-2\} \subset \mathbb{Z}^b$. Then the vector of signed maximal minors of L is $\pm x^{b-1} \operatorname{adj}(\bar{\varphi})$; hence $L^\perp$ is spanned by $x^{b-1} \operatorname{adj}(\bar{\varphi})$ and $[\operatorname{sat} L : L] = \operatorname{cont} \operatorname{adj}(\bar{\varphi})$. Moreover*

$$
\begin{aligned}
\operatorname{cont} \operatorname{adj}(\bar{\varphi}) = 1 \iff & \operatorname{rank}_{\mathbb{F}_p}(m_\varphi) \geqslant b-1 \quad \textit{for all primes } p \\
\iff & \deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1) \leqslant 1 \quad \textit{for all primes } p.
\end{aligned}
$$

Proof. The columns of m_φ are $x^j \varphi$, $0 \leqslant j \leqslant b-1$, so the minors of L (delete row i) are the cofactors $C_{i,b-1}$ of m_φ; i.e. $w = \pm(\text{row } b-1 \text{ of } \operatorname{adj} m_\varphi) = \pm \operatorname{adj}(m_\varphi)^T e_{b-1} =$

$\pm\operatorname{adj}(m_{\bar\varphi})e_{b-1} = \pm x^{b-1}\operatorname{adj}(\bar\varphi)$, using $m_\varphi^T = m_{\bar\varphi}$ and the fact that $\operatorname{adj}(m_{\bar\varphi})$ is multiplication by $\operatorname{adj}(\bar\varphi)$. The content of w is the index. For the equivalences: $\operatorname{adj}(M) \equiv 0 \pmod p$ iff all $(b-1)$-minors of M vanish mod p iff $\operatorname{rank}_{\mathbb{F}_p} M \leqslant b-2$; and $\operatorname{rank}_{\mathbb{F}_p} m_\varphi = \operatorname{rank}_{\mathbb{F}_p} m_{\bar\varphi} = \dim_{\mathbb{F}_p}\big(\varphi\cdot\mathbb{F}_p[x]/(x^b-1)\big) = b - \deg\gcd_{\mathbb{F}_p}(\varphi, x^b-1)$, since the image of multiplication by φ in the principal ideal ring $\mathbb{F}_p[x]/(x^b-1)$ is the ideal generated by $\gcd(\varphi, x^b-1)$. $\square$

Theorem 16 (normal vector)**.** *Let $Q \geqslant 8K_s\rho$ and let the level data be as in Definition 13. Then $L_s(Q,\rho)$ has rank $d-1$ and $L_s(Q,\rho) \subseteq R(u)$, where*

$$u := \tilde u_1 \otimes \cdots \otimes \tilde u_s, \qquad u(x_1,\ldots,x_s) = \prod_{k=1}^{s} \tilde u_k(x_k),$$

is a primitive vector of $\mathbb{Z}^d$ with positive coordinates.

Proof. Rank $d-1$: Lemma 9 for the real scale $Q/\rho > K_s$. Primitivity and positivity of u: the coordinates of u are the products $\prod_k \tilde u_{k,i_k}$, positive by Lemma 14, and the content of a tensor product of integer vectors is the product of the contents (Gauss's lemma), so $\operatorname{cont} u = 1$.

Orthogonality. Fix (k,i,m). Since $\langle u, \ell_{k,i,m}\rangle = [x^0]\big(u\,\overline{\Phi_k}\,x_k^{-i}\bar m\big) = [x_k^i m]\big(u\overline{\Phi_k}\big)$, it suffices to show that every monomial of $u\overline{\Phi_k}$ whose $x_1,\ldots,x_{k-1}$-exponents vanish has x_k-exponent $b-1$. Now

$$\begin{aligned} u\overline{\Phi_k} = \frac{Q}{2^k}\prod_{j<k}\Big[\tilde u_j(x_j)(2+x_j^{-1})\Big]\cdot \tilde u_k(x_k)\bar g(x_k)\cdot\prod_{j>k}\tilde u_j(x_j) \\ +\rho\prod_{j<k}\Big[\tilde u_j(x_j)x_j^{-(b-1)}\Big]\cdot\tilde u_k(x_k)\cdot\prod_{j>k}\tilde u_j(x_j), \end{aligned}$$

a sum of two products of univariate factors. Extracting, for each $j<k$, the part with x_j-exponent 0 amounts to replacing the j-th factor by its constant coefficient: $[x_j^0]\tilde u_j(2+x_j^{-1}) = 2\tilde u_{j,0} + \tilde u_{j,1} = c_j$ and $[x_j^0]\tilde u_j x_j^{-(b-1)} = [x_j^0]\tilde u_j x_j = \tilde u_{j,b-1} = a_j$. Hence the relevant part of $u\overline{\Phi_k}$ is

$$\big(\theta_k\bar g(x_k)+\rho_k\big)\tilde u_k(x_k)\prod_{j>k}\tilde u_j = \overline{\varphi_k}\,\tilde u_k\prod_{j>k}\tilde u_j = \pm\frac{\mathrm{N}_k}{\gamma_k}\,x_k^{b-1}\prod_{j>k}\tilde u_j(x_j),$$

by $\overline{\varphi_k}\psi_k = \mathrm{N}_k$. All its monomials have x_k-exponent $b-1$, as required. $\square$

Theorem 17 (index formula)**.** *Under the hypotheses of Theorem 16,*

$$[\,R(u) : L_s(Q,\rho)\,] = \prod_{k=1}^{s}\gamma_k^{b^{s-k}}.$$

In particular $L_s(Q,\rho)$ is primitive if and only if $\gamma_1 = \cdots = \gamma_s = 1$.

Proof. Induction on s. For $s=1$, $L_1(Q,\rho)=\operatorname{span}\{x^i\varphi_1\}$ and the index is γ_1 by Proposition 15. Let $s\geqslant 2$, $d'=b^{s-1}$, and view $\mathbb{Z}^d=\mathbb{Z}^b\otimes\mathbb{Z}^{d'}$ with x_1 in the first factor. The family $k=1$ spans $L^{(1)}\otimes\mathbb{Z}^{d'}$ with $L^{(1)}=\operatorname{span}\{x_1^i\varphi_1\}$, whose orthogonal complement in $\mathbb{R}^b$ is $\mathbb{R}\tilde{u}_1$. For $k\geqslant 2$ write $\Phi_k=(2+x_1)\otimes A_k+x_1^{b-1}\otimes B_k$ with $A_k=\frac{Q}{2^k}(1-x_k)\prod_{2\leqslant j\leqslant k}(2+x_j)$ and $B_k=\rho\prod_{2\leqslant j<k}x_j^{b-1}$ in $R_{s-1}(x_2,\dots,x_s)$. Decompose $\mathbb{R}^b\otimes\mathbb{R}^{d'}=W\oplus W^\perp$ with $W=\tilde{u}_1^\perp\otimes\mathbb{R}^{d'}$ and $W^\perp=\mathbb{R}\tilde{u}_1\otimes\mathbb{R}^{d'}$, and identify $W^\perp$ with $\mathbb{R}^{d'}$ via $\tilde{u}_1/|\tilde{u}_1|_2\otimes y\mapsto y$. The projection of $\ell_{k,i,m}$ ($k\geqslant 2$) onto $W^\perp$ is

$$\frac{1}{|\tilde{u}_1|_2}\big(\langle\tilde{u}_1,2+x_1\rangle A_k+\langle\tilde{u}_1,x_1^{b-1}\rangle B_k\big)x_k^im=\frac{1}{|\tilde{u}_1|_2}\left(c_1A_k+a_1B_k\right)x_k^im,$$

and $c_1A_k+a_1B_k=\frac{c_1Q/2}{2^{k-1}}(1-x_k)\prod_{2\leqslant j\leqslant k}(2+x_j)+a_1\rho\prod_{2\leqslant j<k}x_j^{b-1}$, so these projections form, up to the factor $1/|\tilde{u}_1|_2$, the basis of $L':=L_{s-1}(Q',\rho')$ with $Q'=c_1Q/2$ and $\rho'=a_1\rho$ in the variables $x_2,\dots,x_s$. The level data of L' are $(\theta_2,\rho_2),\dots,(\theta_s,\rho_s)$: indeed $Q'/2=\theta_2$, $\rho'=\rho_2$, and the recursion is the same. By Lemma 14, $Q'/\rho'=(c_1/a_1)(Q/2\rho)\geqslant Q/\rho\geqslant 8K_s\geqslant 8K_{s-1}$, so the induction hypothesis applies to L': its normal vector is $u'=\tilde{u}_2\otimes\cdots\otimes\tilde{u}_s$ and $[R(u'):L']=\prod_{k\geqslant 2}\gamma_k^{b^{s-k}}$. Since L' has rank $d'-1$, the projection onto $W^\perp$ is injective on $L_{\geqslant 2}:=\operatorname{span}\{\ell_{k,i,m}:k\geqslant 2\}$; hence $L_s(Q,\rho)\cap W=L^{(1)}\otimes\mathbb{Z}^{d'}$, the projection of $L_s(Q,\rho)$ onto $W^\perp$ is $L'/|\tilde{u}_1|_2$, and

$$\operatorname{covol}L_s(Q,\rho)=\operatorname{covol}(L^{(1)}\otimes\mathbb{Z}^{d'})\cdot\operatorname{covol}\big(L'/|\tilde{u}_1|_2\big)=\operatorname{covol}(L^{(1)})^{d'}\cdot\frac{\operatorname{covol}L'}{|\tilde{u}_1|_2^{d'-1}},$$

where $\operatorname{covol}(L^{(1)}\otimes\mathbb{Z}^{d'})=\operatorname{covol}(L^{(1)})^{d'}$ because $L^{(1)}\otimes\mathbb{Z}^{d'}$ is the orthogonal direct sum of the d' copies $L^{(1)}\otimes e_m$. As $u=\tilde{u}_1\otimes u'$ is primitive, $\operatorname{covol}R(u)=|u|_2=|\tilde{u}_1|_2|u'|_2$, so

$$[R(u):L_s(Q,\rho)]=\frac{\operatorname{covol}L_s(Q,\rho)}{\operatorname{covol}R(u)}=\Big(\frac{\operatorname{covol}L^{(1)}}{|\tilde{u}_1|_2}\Big)^{d'}\frac{\operatorname{covol}L'}{|u'|_2}=\gamma_1^{d'}\prod_{k\geqslant 2}\gamma_k^{b^{s-k}}.\qquad\square$$

Proposition 18 (exceptional primes)**.** *Let $n_b=\prod_{0<i<j<b}\operatorname{Res}_z\big(\frac{z^b-1}{z-1},1+z^i+z^j\big)$ and let P_b be the set of odd primes dividing $b(2^b+1)n_b$. Then $n_b\neq 0$, so P_b is finite, and for $\varphi=\theta g+\rho$ with $\theta,\rho\in\mathbb{Z}$ and a prime p:*

(a) if $p\mid\gcd(\theta,\rho)$ then $\deg\gcd_{\mathbb{F}_p}(\varphi,x^b-1)=b$;

(b) if $p\mid\theta$ and $p\nmid\rho$ then $\deg\gcd_{\mathbb{F}_p}(\varphi,x^b-1)=0$;

(c) if $p=2$ and ρ is odd then $\deg\gcd_{\mathbb{F}_2}(\varphi,x^b-1)=0$;

(d) if p is odd, $p\notin P_b$ and $p\nmid\gcd(\theta,\rho)$, then $\deg\gcd_{\mathbb{F}_p}(\varphi,x^b-1)\leqslant 1$.

Proof. For roots of unity, $1+\zeta_1+\zeta_2=0$ forces $\{\zeta_1,\zeta_2\}=\{\omega,\bar{\omega}\}$, which is impossible in μ_b because $3\nmid b$; hence every factor of n_b is a nonzero integer. (a) and (b) are clear: $\varphi\equiv 0$, respectively a nonzero constant, mod p. (c): if θ is even, $\varphi\equiv\rho\equiv 1$; if θ is odd, $\varphi\equiv g+1\equiv 1+x+x^2\pmod 2$, whose roots are the primitive cube roots of unity, which are not b-th roots of unity because $3 \nmid b$. (d): Let $\mathbb{F} = \overline{\mathbb{F}}_p$. Since $p \nmid b$, $x^b - 1$ is separable over $\mathbb{F}$; fix a prime $\mathfrak{p} \mid p$ of $\mathbb{Z}[\zeta_b]$; reduction modulo $\mathfrak{p}$ maps μ_b bijectively onto the roots of $x^b - 1$ in $\mathbb{F}$. As $x^b - 1$ is squarefree over $\mathbb{F}$, $\deg \gcd_{\mathbb{F}_p}(\varphi, x^b - 1)$ is the number of distinct common roots, so suppose $\zeta_1 \neq \zeta_2$ are common roots. If $p \mid \theta$ then $p \nmid \rho$ and $\varphi \equiv \rho$ has no roots. So $p \nmid \theta$ and $g(\zeta_1) = g(\zeta_2) = -\rho/\theta$; since $g(a) - g(c) = -(a-c)(1+a+c)$, we get $1 + \zeta_1 + \zeta_2 = 0$ in $\mathbb{F}$. If $\zeta_1, \zeta_2 \neq 1$, lifting to $\tilde{\zeta}_i \in \mu_b$ gives $\mathfrak{p} \mid (1 + \tilde{\zeta}_1 + \tilde{\zeta}_2)$; writing $\tilde{\zeta}_1 = \eta^i$, $\tilde{\zeta}_2 = \eta^j$ for a primitive $\eta \in \mu_b$, the algebraic integer $1 + \eta^i + \eta^j$ divides $\mathrm{Res}_z(\frac{z^b-1}{z-1}, 1 + z^i + z^j)$, so $p \mid n_b$, a contradiction. If $\zeta_1 = 1$, then $\zeta_2 = -2$, so $(-2)^b = 1$ in $\mathbb{F}$, i.e. $p \mid 2^b + 1$, again a contradiction. □

Theorem 19 (first level)**.** *Let $b \geqslant 5$ be odd, $3 \nmid b$, and let $q \geqslant 4K_1$ be an integer with $\mathrm{rad}(P_b) \mid q$. Then $L_1(2q)$ is primitive. Its normal vector $\tilde{u}_1 = \pm x^{b-1}\,\mathrm{adj}(\bar{\varphi}_1)$, $\varphi_1 = qg + 1$, has positive coordinates forming a $(2q - K_1)$-dissociated set with $\max \tilde{u}_1 \leqslant (2q)^{b-1}\Delta_{b,1}(1 + 5K_1/q)$, and for $2^l \leqslant 2q - K_1$ the lift $\{2^j \tilde{u}_{1,i}\}$ is a dissociated set of size lb with $f(lb) \leqslant \max \tilde{u}_1 / 2^{l(b-1)}$.*

Proof. By Proposition 15 it suffices to check $\deg \gcd_{\mathbb{F}_p}(\varphi_1, x^b - 1) \leqslant 1$ for all p. For odd $p \mid q$ use (b); for odd $p \nmid q$ we have $p \notin P_b$ and $\gcd(q, 1) = 1$, so (d) applies; $p = 2$ is (c). The remaining assertions are Lemmas 9 and 10 with $K = K_1$, $Q = 2q$, together with (2). □

Example 20. Let $b = 5$ and $Q = 16$, so $\varphi_1 = 8g + 1 = 17 - 8x - 8x^2$ and $\overline{\varphi_1} = 17 - 8x^3 - 8x^4$. One finds

$$\psi_1 = \mathrm{adj}(\overline{\varphi_1}) = 62017 + 48704x + 49792x^2 + 53704x^3 + 52104x^4,$$

$\mathrm{N}_1 = 266321 = 11^2 \cdot 31 \cdot 71$, with $\psi_1 \overline{\varphi_1} = 266321$ in R_5, $\gamma_1 = 1$, and $\tilde{u}_1 = x^4 \psi_1 = (48704,\ 49792,\ 53704,\ 52104,\ 62017)$. Here $K_1 = 424/55$, so Lemma 9 guarantees that the coordinates are 8-dissociated (they are in fact 16-dissociated but not 17-dissociated, by brute force), and $\max \tilde{u}_1 / Q^4 = 62017/65536 \approx 0.946$, compared with $\Delta_{5,1} = 11/16$; the ratio tends to $\Delta_{5,1}$ as $Q \to \infty$, e.g. it is 0.6906 for $Q = 8200$. The lift with $l = 3$ is the dissociated 15-element set $\{\tilde{u}_{1,i}, 2\tilde{u}_{1,i}, 4\tilde{u}_{1,i}\}$. Finally $a_1 = 62017$ and $c_1 = 2 \cdot 48704 + 49792 = 147200$ are coprime, so the second level can be built on top of this one.

5 Explicit dissociated sets

Theorem 21. *Let $b \geqslant 5$ be odd with $3 \nmid b$, $s \geqslant 1$, $d = b^s$, $Q \in 2^s\mathbb{Z}$ with $Q \geqslant 8K_s$, and suppose that the level data of Definition 13 (with $\rho = 1$) satisfy $\gamma_1 = \cdots = \gamma_s = 1$. Then $u = \tilde{u}_1 \otimes \cdots \otimes \tilde{u}_s$ is a vector of d positive integers whose coordinates are $(Q - K_s)$-dissociated, and*

$$\max u = \prod_{k=1}^{s} \max_i \tilde{u}_{k,i} \;\leqslant\; Q^{d-1}\Delta_{b,s}\Big(1 + \frac{10K_s}{Q}\Big).$$

For every $l \geqslant 1$ with $2^l \leqslant Q - K_s$, the set $A = \{2^j u_i : 0 \leqslant j < l,\ 1 \leqslant i \leqslant d\}$ is dissociated, $|A| = ld$, and

$$f(ld) \;\leqslant\; \frac{\max u}{2^{l(d-1)}} \;\leqslant\; \Delta_{b,s}\Big(1 + \frac{10K_s}{Q}\Big)\Big(\frac{Q}{2^l}\Big)^{d-1}.$$

Proof. Theorems 16 and 17 give $L_s(Q) = R(u)$; Lemma 9 with $K = K_s$ (Proposition 11) gives $R(u) \cap (-Q - K_s, Q - K_s)^d = \{0\}$; Lemma 10(ii) bounds $\max u$; the product formula holds because all $\tilde{u}_{k,i} > 0$; finally apply (2). $\square$

The sizes ld in Theorem 21 form a sparse set. The following classical observation, used by Conway and Guy and by Bohman [4] to pass from one size to all larger sizes, fills the gaps at no cost.

Lemma 22 (monotonicity)**.** *If A is dissociated with $|A| = n$, then $2A \cup \{1\}$ is dissociated with $n + 1$ elements and maximum $2 \max A$. Consequently $f(n+1) \leqslant f(n)$ for all $n \geqslant 1$, and for every $m \geqslant n$ the explicit set $A_m = 2^{m-n} A \cup \{1, 2, \ldots, 2^{m-n-1}\}$ is dissociated with $|A_m| = m$ and $\max A_m / 2^m = \max A / 2^n$.*

Proof. The elements of $2A$ are even and 1 is odd, so $2A \cup \{1\}$ has $n + 1$ elements. If two subsets of $2A \cup \{1\}$ have the same sum, the two sums have the same parity, so either both subsets contain 1 or neither does; removing 1 and halving gives two subsets of A with the same sum, hence the same subsets. The maximum is $2 \max A$ because $\max A \geqslant 1$, and $2 \max A / 2^n = \max A / 2^{n-1}$; taking the infimum over A gives $f(n+1) \leqslant f(n)$, and iterating the construction $m - n$ times gives A_m. $\square$

Corollary 23. *Table 1 and Lemma 22 give explicit dissociated sets A of every size n with*

$$\begin{aligned} &\max A < 0.158523 \cdot 2^n && \text{for every } n \geqslant 43\,923, \\ &\max A < 0.132535 \cdot 2^n && \text{for every } n \geqslant 1\,199\,562, \\ &\max A < 0.103036 \cdot 2^n && \text{for every } n \geqslant 3\,841\,966. \end{aligned}$$

In particular $f(n) \leqslant 0.206072$ for all $n \geqslant 3\,841\,966$.

Proof. The rows $(b, s) = (11, 3)$, $(13, 4)$ and $(17, 4)$ of Table 1 provide dissociated sets of sizes $43\,923$, $1\,199\,562$ and $3\,841\,966$ with $\max A / 2^{|A|}$ below 0.158523, 0.132535 and 0.103036 respectively (Theorem 21 with the certificates of Section 6); apply Lemma 22. $\square$

5.1 An arithmetic progression of good scalings

Write $Q = 2^{s+1} t$ with $t \in \mathbb{Z}_{>0}$. Run the recursion of Definition 13 formally, without dividing by contents and without sign normalisation: $\hat{\theta}_1 = 2^s t$, $\hat{\rho}_1 = 1$, $\hat{\varphi}_k = \hat{\theta}_k g + \hat{\rho}_k$, $\hat{\psi}_k = \operatorname{adj}(\overline{\hat{\varphi}_k})$, $\hat{a}_k = \hat{\psi}_{k,0}$, $\hat{c}_k = 2\hat{\psi}_{k,1} + \hat{\psi}_{k,2}$, $\hat{\theta}_{k+1} = 2^{s-k} t\, \hat{c}_1 \cdots \hat{c}_k$, $\hat{\rho}_{k+1} = \hat{a}_1 \cdots \hat{a}_k$. All of these are polynomials in t with integer coefficients (adjugates of matrices with entries in $\mathbb{Z}[t]$). If $\gamma_1 = \cdots = \gamma_k = 1$ at a given t, then $\psi_j = \hat{\psi}_j$ for $j \leqslant k$: the signs ε_j enter θ_j and ρ_j through the same product, so $\varphi_j = \pm \hat{\varphi}_j$, and adj is invariant under $\varphi \mapsto -\varphi$ because $b - 1$ is even. Hence $a_j = \pm \hat{a}_j(t)$ and $c_j = \pm \hat{c}_j(t)$ with a common sign.

Definition 24 (the coprimality condition $\boldsymbol{H(b,s)}$)**.** We say that $H(b, s)$ holds if for each $1 \leqslant k \leqslant s - 1$ the polynomials $\hat{a}_k, \hat{c}_k \in \mathbb{Z}[t]$ are coprime in $\mathbb{Q}[t]$; equivalently, $R_k := \operatorname{Res}_t(\hat{a}_k, \hat{c}_k) \in \mathbb{Z} \setminus \{0\}$.

Theorem 25. *Assume $H(b,s)$ and put $M = \operatorname{rad}(P_b)\prod_{k<s}|R_k|$. If $t \equiv 0 \pmod M$ and $Q = 2^{s+1}t \geqslant 8K_s$, then $\gamma_1 = \cdots = \gamma_s = 1$; that is, $L_s(Q)$ is primitive and Theorem 21 applies.*

Proof. We show by induction on k that level k is primitive, i.e. $\deg \gcd_{\mathbb{F}_p}(\varphi_k, x^b - 1) \leqslant 1$ for every prime p (Proposition 15), assuming that levels $j < k$ are primitive. Recall $\theta_k = 2^{s+1-k}t\, c_1 \cdots c_{k-1}$ and $\rho_k = a_1 \cdots a_{k-1}$.

Claim: for every prime $p \mid 2t$ and every $j < k$, $p \nmid a_j$. Indeed $p \mid \theta_j$ (for $p = 2$ because 2^{s+1-j} is even), so $\varphi_j \equiv \rho_j$ and $\psi_j \equiv \operatorname{adj}(\rho_j \cdot 1) = \rho_j^{b-1}$ (a constant) mod p; as $\gamma_j = 1$, $a_j = \pm\psi_{j,0} \equiv \pm\rho_j^{b-1}$, and $p \nmid \rho_j = a_1 \cdots a_{j-1}$ by induction on j.

Case $p \mid 2t$. Then $p \mid \theta_k$ and $p \nmid \rho_k$ by the claim, so Proposition 18(b) applies.

Case p odd, $p \nmid t$. Then $p \notin P_b$ because $\operatorname{rad}(P_b) \mid t$. By Proposition 18(d) it suffices to show $p \nmid \gcd(\theta_k, \rho_k)$. Suppose $p \mid \theta_k$ and $p \mid \rho_k$; as $p \nmid 2t$, $p \mid c_j$ for some $j < k$, and $p \mid a_{j'}$ for some $j' < k$. Choose such a pair with j' minimal. If $j' > j$: $p \mid c_j \mid \theta_{j'}$, so as in the claim $a_{j'} \equiv \pm\rho_{j'}^{b-1} \bmod p$, whence $p \mid \rho_{j'}$ and $p \mid a_{j''}$ for some $j'' < j'$, contradicting minimality. If $j' < j$: $p \mid a_{j'} \mid \rho_j$, and $p \nmid \theta_j$ (else $\varphi_j \equiv 0$ and level j would not be primitive), so $\varphi_j \equiv \theta_j g$ and $\psi_j \equiv \theta_j^{b-1}\operatorname{adj}(\bar{g}) \bmod p$. The circulant $m_{\bar{g}}$ has the simple eigenvalue 0 with eigenvector $\mathbf{1}_b$, its other eigenvalues $\bar{g}(\zeta)$, $\zeta \neq 1$, being nonzero mod p (since $g(\zeta) = 0$ forces $\zeta = -2$, i.e. $p \mid 2^b + 1$); hence $\operatorname{adj}(m_{\bar{g}}) = \prod_{\zeta\neq 1}\bar{g}(\zeta) \cdot \frac{1}{b}\mathbf{1}_b\mathbf{1}_b^T = \frac{2^b+1}{3}\mathbf{1}_b\mathbf{1}_b^T$, because $\prod_{\zeta\neq1}(1-\zeta) = b$ and $\prod_{\zeta\neq 1}(2+\zeta) = (2^b+1)/3$. Thus $\psi_j \equiv \theta_j^{b-1}\frac{2^b+1}{3}\sum_i x^i$ and $c_j \equiv \pm\theta_j^{b-1}(2^b+1) \not\equiv 0$ mod p, contradicting $p \mid c_j$. Hence $j' = j$: $p \mid \gcd(a_j, c_j) = \gcd(\hat{a}_j(t), \hat{c}_j(t))$, and since $\hat{a}_j, \hat{c}_j$ are coprime in $\mathbb{Q}[t]$ there are $A, B \in \mathbb{Z}[t]$ with $A\hat{a}_j + B\hat{c}_j = R_j$, so $p \mid R_j \mid M \mid t$, a contradiction. $\square$

Corollary 26. *If $H(b,s)$ holds, then $\limsup_{l\to\infty} f(l\,b^s) \leqslant \Delta_{b,s}$.*

Proof. For large l choose $t \equiv 0 \pmod M$ with $Q = 2^{s+1}t \in [2^l + K_s,\, 2^l + K_s + 2^{s+1}M]$. Theorems 25 and 21 give $f(ld) \leqslant \Delta_{b,s}(1 + 10K_s/Q)(Q/2^l)^{d-1} \to \Delta_{b,s}$ as $l \to \infty$. $\square$

Corollary 27. *$H(b,s)$ holds for the eight pairs $(5,2)$, $(7,2)$, $(11,2)$, $(13,2)$, $(5,3)$, $(7,3)$, $(11,3)$ and $(13,3)$ (see Section 6). Hence, unconditionally,*

$$\limsup_{l\to\infty} f(1331\,l) \leqslant \Delta_{11,3} < 0.31617, \qquad \limsup_{l\to\infty} f(2197\,l) \leqslant \Delta_{13,3} < 0.30299,$$

and, since f is nonincreasing (Lemma 22), $\lim_{n\to\infty} f(n) \leqslant \Delta_{13,3}$; in particular $f(n) < 0.303$ for all sufficiently large n.

Corollary 27 uses only the certificates $H(11,3)$ and $H(13,3)$ of Section 6, item 6, and no large adjugate computation; the explicit sets of Table 1 give the sharper effective statement $f(n) \leqslant 0.206072$ for all $n \geqslant 3\,841\,966$ (Corollary 23).

Conjecture 28. $H(b,s)$ holds for all odd $b \geqslant 5$ with $3 \nmid b$ and all $s \geqslant 1$.

Theorem 29 (conditional rate)**.** *Assume Conjecture 28. Then there is an absolute constant $c > 0$ (one may take $c = 1/20$) such that $f(n) \leqslant n^{-c/\log\log n}$ for infinitely many n.*

We only sketch the proof. The omitted details are height estimates for the adjugates $\hat{\psi}_k$ and for the resultants R_k as polynomials in t; the numerical value of c depends on them, the existence of some $c > 0$ does not.

Sketch. For $s \geqslant 4$ let $b = b(s)$ be the least odd integer $\geqslant 5$, not divisible by 3, with $2^{-b}b^{s-1} \leqslant \frac{1}{2}$; then $b \leqslant 2s\log_2 s$ and $\Delta_{b,s} \leqslant e\,(2/3)^s$ by (1). Put $d = b^s$, so that $s \geqslant \log d/\log(2s\log_2 s) \geqslant \log d/(2\log\log d)$ for large s. By Theorem 25 there is a good Q in every interval $[X, X + 2^{s+1}M]$ with $X \geqslant 8K_s$, so we need to bound $\log M$. The polynomials $\hat{a}_k, \hat{c}_k$ have degree $D_k \leqslant b^k \leqslant d$ in t and, by Hadamard's inequality for the adjugate and trivial bounds for products, height $\exp(O(d\,s\log b))$; hence $\log|R_k| \leqslant 2D_k\log(\text{height}) + O(D_k\log D_k) = O(sd^2\log b)$ and $\log M = O(s^2d^2\log b)$. Taking $l = \lceil\log_2(2^{s+1}M+8K_s)\rceil+\lceil\log_2(200d)\rceil$ we get $(Q/2^l)^{d-1}(1+10K_s/Q) \leqslant e^{1/100}$ and $n = ld \leqslant d^4$ for large d, so $\log d \geqslant \frac{1}{4}\log n$ and $\log\log d \leqslant \log\log n$. Therefore $f(n) \leqslant 3(2/3)^s \leqslant 3\exp\bigl(-\log(3/2)\log d/(2\log\log d)\bigr) \leqslant 3\,n^{-\log(3/2)/(8\log\log n)} \leqslant n^{-c/\log\log n}$ with $c = 1/20$, for all large s. $\square$

Corollary 30. *Assume Conjecture 28. Then $f(n) \leqslant n^{-c'/\log\log n}$ for all sufficiently large n, with $c' = 1/50$.*

Proof. For each large s the proof of Theorem 29 produced a size $n_s = l_sd_s$ with $d_s = b(s)^s$, $n_s \leqslant d_s^4$ and $f(n_s) \leqslant 3(2/3)^s$. Given a large n, let s be maximal with $n_s \leqslant n$. Then $n < n_{s+1} \leqslant d_{s+1}^4$, so $\log n < 4(s+1)\log b(s+1) \leqslant 8(s+1)\log(s+1) \leqslant 16\,s\log s$ for $s \geqslant 3$, using $b(s+1) \leqslant 2(s+1)\log_2(s+1)$. Since $5^s \leqslant d_s \leqslant n$ we have $s < \log_2 n$, hence $\log s \leqslant 1.1\log\log n$ and $s > \log n/(18\log\log n)$ for large n. By Lemma 22,

$$f(n) \leqslant f(n_s) \leqslant 3\,(2/3)^s \leqslant 3\exp\Bigl(-\frac{\log(3/2)\,\log n}{18\log\log n}\Bigr) \leqslant n^{-1/(50\log\log n)}$$

for all large n, because $\log(3/2)/18 > 0.0225 > 1/50$. $\square$

Remark 31. If good scalings Q had positive density in $[X, 2X]$, as the heuristics and all our experiments suggest, one could take $l = \log_2 d + O(\log\log d)$ and the exponent would become $c = \log(3/2) - o(1)$, the value predicted by $\Delta_{b,s} \approx (2/3)^s$ with $s \approx \log d/\log\log d$.

6 Computations and certificates

All computations use exact integer arithmetic (Python, with `sympy` for polynomial manipulations); the adjugate elements are computed by fraction-free Bareiss elimination, and every ψ_k is checked against the identity $\psi_k\overline{\varphi_k} = \mathrm{N}_k$. The code and the certificate data are available with this paper (see the Acknowledgements).

1. *Correctness checks.* For $(b,s) \in \{(5,1),(5,2),(3,2),(7,2)\}$ and several Q we built the basis of $L_s(Q)$ explicitly, computed its vector of maximal minors directly, and verified that it equals $\pm\prod_k \gamma_k^{b^{s-k}} \cdot u$ with $u = \otimes_k\tilde{u}_k$, as predicted by Theorems 16 and 17 (for instance $(b,s) = (3,2)$, $Q = 12$ gives $\gamma = (19,169)$ and index $19^3 \cdot 169$; the case $3 \mid b$ is

not covered by our theorems but the algebra of Section 4 still applies). The invariants h_s, $\mathrm{covol}(\Gamma_s)$ and $\Delta_{b,s}$ of Proposition 8 were verified numerically, and for $(b,s)=(3,2)$ the conditions (A1)–(A2) were checked by exhaustive enumeration of $5.7\cdot 10^6$ coefficient vectors.

2. *Brute-force dissociation.* The lifted sets of Theorem 19 with $b=5$, $l\in\{3,4,5\}$ (for $Q=16,24,40$) and with $b=7$, $l=3$ (for $Q=24$), of sizes 15, 20, 25 and 21, as well as the 25-element vectors $u=\tilde{u}_1\otimes\tilde{u}_2\in\mathbb{Z}^{25}$ of Definition 13 for $(b,s)=(5,2)$ and $Q\in\{108,116,124\}$ (scalings below the threshold $8K_2$ of Theorem 21, which therefore does not cover them), were verified to be dissociated by a meet-in-the-middle enumeration of all signed sums.

3. *Sharpness of Lemma 9.* For $b=5$ and $Q\in\{16,24,40\}$ the coordinates of $\tilde{u}_1$ are in fact exactly Q-dissociated, whereas the lemma guarantees $Q-K_1\approx Q-7.7$. We do not know whether $L_s(Q)\cap(-Q,Q)^d=\{0\}$ holds in general; if it did, one could take $Q=2^l$ whenever 2^l is a good scaling, and the factor $(Q/2^l)^{d-1}$ in Theorem 21 would disappear.

4. *Level criterion.* For $b\in\{5,7,9,11,13\}$ and $8\leqslant q<40$ the criterion of Proposition 15, evaluated by factoring $\mathrm{Res}(\varphi_1,x^b-1)$ and computing polynomial gcds mod p, agreed with the directly computed index in all 160 cases. For $3\mid b$ the first level is never primitive: $\varphi_1(\omega)=3q+1$ for both primitive cube roots of unity ω, so every prime factor of $3q+1$ is exceptional; this is why we assume $3\nmid b$.

5. *Theorem 19.* $P_5=\{3,5,11\}$, $P_7=\{3,7,43\}$, $P_{11}=\{3,11,23,683\}$; for $q=m\cdot\mathrm{rad}(P_b)$, $1\leqslant m\leqslant 40$, the first level was primitive in all 120 cases. The level-one polynomials satisfy $\hat{a}_1(q)=\mathrm{N}(q)-\frac{q}{b}\mathrm{N}'(q)$ with $\mathrm{N}(q)=\prod_{\zeta\neq 1}(1+qg(\zeta))$, and $\mathrm{Res}_q(\hat{a}_1,\hat{c}_1)$ equals $2\cdot 11\cdot 19$, $2^3\cdot 7\cdot 43\cdot 379$, $2\cdot 3^5\cdot 5^2\cdot 23^3\cdot 683\cdot 14797$ and $3^3\cdot 13\cdot 53^3\cdot 61\cdot 2731\cdot 626592619$ for $b=5,7,11,13$; the primes 19 (for $b=5$) and 7 (for $b=7$) are exactly the second-level failures observed at $Q=20$ and $Q=12$.

6. *Certification of $H(b,s)$.* If a non-constant $f\in\mathbb{Z}[t]$ divided both $\hat{a}_k$ and $\hat{c}_k$, then $f(t)\mid\gcd(\hat{a}_k(t),\hat{c}_k(t))$ for all $t\in\mathbb{Z}$, so $\gcd(\hat{a}_k(t),\hat{c}_k(t))=1$ would force $|f(t)|=1$, which is possible for at most $2\deg f\leqslant 2D_k$ integers t, where $D_k=(b-1)\deg\hat{\theta}_k$ bounds $\deg\hat{a}_k$. (Indeed $\deg\hat{\rho}_k<\deg\hat{\theta}_k$ by induction, so the leading form of $\mathrm{adj}(\overline{\hat{\varphi}_k})$ in t is $\hat{\theta}_k^{b-1}\,\mathrm{adj}(\bar{g})=\hat{\theta}_k^{b-1}\frac{2^b+1}{3}\sum_i x^i$, whence $\deg\hat{a}_k=\deg\hat{c}_k=(b-1)\deg\hat{\theta}_k$ and $\deg\hat{\rho}_{k+1}=\deg\hat{\theta}_{k+1}-1$.) We exhibited more than $2D_k$ integers t (with levels $\leqslant k$ primitive) and $\gcd(a_k,c_k)=1$, for each $k<s$ and each (b,s) in Corollary 27; e.g. for $(11,3)$, $D_1=10$, $D_2=110$, and 246, respectively 221, good values of t occur among the first 426. For $(5,2)$ and $(7,2)$ we also computed $\mathrm{Res}_t(\hat{a}_2,\hat{c}_2)\neq 0$ directly (297 and 1319 decimal digits).

7. *The table.* For each (b,s) in Table 1 we chose $l=\lceil\log_2(200(d-1)K_s)\rceil$, so that $(Q/2^l)^{d-1}\leqslant e^{1/200}$, and the least t with $Q=2^{s+1}t\geqslant 2^l+K_s$ and $\gamma_1=\cdots=\gamma_s=1$; at most three values of t were needed in every case. For $(17,4)$ the four adjugates involve

integers of about $3 \cdot 10^6$ bits. The column $f(n)$ is the exact rational $\max u/2^{l(d-1)}$ of Theorem 21, rounded up.

b	s	$d = b^s$	l	$n = ld$	Q	$f(n) \leqslant$	$\max A/2^n \leqslant$	$\Delta_{b,s}$
5	1	5	13	65	8200	0.690649	0.345325	0.68750
7	1	7	15	105	32784	0.674014	0.337007	0.67188
11	1	11	16	176	65572	0.670804	0.335402	0.66699
13	1	13	17	221	131120	0.669767	0.334884	0.66675
5	2	25	19	475	524392	0.537132	0.268566	0.53457
7	2	49	22	1078	4194584	0.474515	0.237258	0.47299
11	2	121	25	3025	33555528	0.448812	0.224406	0.44706
13	2	169	26	4394	67110656	0.447207	0.223604	0.44520
5	3	125	25	3125	33555168	0.771242	0.385621	0.76915
7	3	343	28	9604	268438368	0.463429	0.231715	0.46171
11	3	1331	33	43923	8589952400	0.317045	0.158523	0.31617
13	3	2197	34	74698	17179903936	0.304338	0.152169	0.30299
17	3	4913	37	181781	137439055184	0.298073	0.149037	0.29699
13	4	28561	42	1199562	4398047063424	0.265069	0.132535	0.26412
17	4	83521	46	3841966	70368746289408	0.206072	0.103036	0.20556

Table 1: Certified explicit dissociated sets A of size $n = ld$ (Theorem 21). The set A is the lift of $u = \otimes_k \tilde{u}_k$; u is determined by (b, s, Q) through Definition 13, and the certificate is $\gamma_1 = \cdots = \gamma_s = 1$. The previous explicit record is $\max A \leqslant 0.22002 \cdot 2^n$ [4].

Table 1 is read as follows. In every row the column $f(n) \leqslant$ is within 0.6% of $\Delta_{b,s}$ (the largest ratio, about 1.006, occurs at $(11, 1)$), so the losses from the finite l and from the factor $1 + 10K_s/Q$ are negligible, and what a pair (b, s) can achieve is essentially $\Delta_{b,s}$. Since the bound of Theorem 21 is never below $\Delta_{b,s} > (2/3)^s$, no pair with $s \leqslant 2$ can certify $\max A/2^n < \frac{1}{2}(2/3)^2 = 2/9$, let alone Bohman's 0.22002; for $s = 3$ one needs $2^{-b}b^2$ small, and $(7, 3)$ still gives 0.2317 while $(11, 3)$ is the first pair below 0.22002. Increasing s further requires b to grow like $s \log_2 s$ to keep $(1 + 2^{-b})^{(b^s-1)/(b-1)}$ bounded, which is the mechanism behind Theorem 29; the price is the dimension $d = b^s$ of the vector and the height of the adjugates, about $3 \cdot 10^6$ bits for $(17, 4)$. The certificate itself is always s adjugates of $b \times b$ integer matrices, and each row was obtained with at most three trial values of t. No row is claimed to be optimal for its size; by Lemma 22, every row is also an upper bound for $f(n)$ at all larger n (Corollary 23).

7 Remarks and open problems

7.1 Sizes reached by the construction

Our sizes are multiples ld of $d = b^s$ with $l \geqslant \log_2(dK_s)$; since numerically $K_1 \approx 0.26\, b^2$ and hence, by Proposition 11, $K_s = O(b^2 d)$, they are of the form $n \approx 2d \log_2 d$ with

$d = b^s$, a sparse set. Lemma 22 fills the gaps between consecutive sizes without any loss in $\max A/2^{|A|}$, and it is the only cheap way we know: three natural ways to reach other sizes *inside* the construction all lose a factor exponential in the number of added elements. (i) If A is dissociated with sum S, then $A \cup \{S+1\}$ is dissociated, but this multiplies f by about $|A|/2$. (ii) Mixed lifts $\bigcup_i \{2^j u_i : j < l_i\}$ are dissociated as soon as u is $2^{\max_i l_i}$-dissociated, but if r coordinates use $l+1$ doublings the bound becomes $f(ld+r) \leqslant 2^{d-r}\Delta(1+o(1))$. (iii) A lift $B \cdot U$ by a dissociated set B with $\sum B < Q$ has $|B||U|$ elements and loses the factor $(\sum B/2^{|B|})^{|U|-1} \geqslant 1$, with equality only for $B = \{1, 2, \ldots, 2^{l-1}\}$. Padding the lattice itself, from V_d to V_{d+1} via $\Lambda^+ = \{(\lambda + kv, -kh)\}$, gives an admissible pair but multiplies Δ by the height h. Two consequences of Lemma 22 deserve to be made explicit. First, together with the non-effective disproof it shows that $f(n) \to 0$ along all n, not only along a subsequence. Second, the limit $\lim_{n\to\infty} f(n)$ exists, and each entry of Table 1 is an explicit upper bound for it; the effective content of this paper is the rate at which these explicit bounds decrease (Theorem 29 and Corollary 30).

7.2 The exponent and heights of admissible pairs

Proposition 32. *Let (Λ, v) be admissible in $\mathbb{R}^b$ with height h and* $\operatorname{covol}(\Gamma) = 1 + \eta$. *Its s-fold tensor power (Proposition 6) is admissible in dimension $d = b^s$ with $\Delta \leqslant (1+\eta)^{(b^s-1)/(b-1)} h^{-s}$. Conversely $\Delta \geqslant 1/\sqrt{b}$ for every admissible pair in $\mathbb{R}^b$, so $h = \operatorname{covol}(\Gamma)/\Delta \leqslant (1+\eta)\sqrt{b}$.*

Proof. The first part is Proposition 6 iterated. For the second, (A1) says that Λ is a packing lattice for $\frac{1}{2}(-1,1)^b \cap V_b$, so Minkowski's theorem gives

$$\operatorname{covol}(\Lambda) \geqslant 2^{-(b-1)} \operatorname{vol}_{b-1}\big((-1,1)^b \cap V_b\big) \geqslant 1,$$

the last inequality by Vaaler's cube slicing theorem [14]; hence $\Delta = \operatorname{covol}(\Lambda)/\sqrt{b} \geqslant b^{-1/2}$. ◻

In the base pair $h = 3/2$ and $\eta = 2^{-b}$; the exponent $c = \log(3/2)$ in $n^{-c/\log\log n}$ is $\log h$. An admissible pair in dimension b with $h \geqslant b^\alpha$ and $\eta \leqslant b^{-Cs}$ would give $\Delta \leqslant 2b^{-\alpha s} = 2d^{-\alpha}$, i.e. (given a primitivisation, and using Lemma 22 to pass to all n) $f(n) \leqslant n^{-\alpha+o(1)}$, and by the proposition $\alpha \leqslant \frac{1}{2}$ is the natural limit, matching the Erdős–Moser barrier $f(n) \gg n^{-1/2}$. We therefore pose:

Problem 33. Determine $h^*(b) = \sup\{h : (\Lambda, v) \text{ admissible in } \mathbb{R}^b,\ \operatorname{covol}(\Gamma) \leqslant 1 + 2^{-b}\}$. Is $h^*(b) \to \infty$? Is $h^*(b) \geqslant b^\alpha$ for some $\alpha > 0$?

Any improvement of the height 3/2, even to another constant, improves the constant c; by the proposition the question is equivalent to the existence of lattice packings of the central hyperplane section of the cube with density $\gg b^{\alpha-1/2}$ that also satisfy the "real-t" condition (A2).

7.3 Siegel's lemma

As noted in [2], and observed earlier by Schinzel and Aliev [1], if C_d denotes the optimal constant in Siegel's lemma for one equation in d unknowns (so that every nonzero $a \in \mathbb{Z}^d$ has a nonzero integer relation x with $|x|_\infty^{d-1} \leqslant C_d|a|_\infty$; see [5]), then $f(d) \geqslant C_d^{-1}$. Each vector u of Theorem 21 is an explicit integer vector of height at most $Q^{d-1}\Delta_{b,s}(1+10K_s/Q)$ all of whose nonzero integer relations c satisfy $|c|_\infty \geqslant Q - K_s$, so it gives the explicit lower bound $C_d \geqslant (1-K_s/Q)^{d-1}/\big(\Delta_{b,s}(1+10K_s/Q)\big)$ for $d = b^s$; for example $C_{1331} \geqslant 3.15$ and $C_{83521} \geqslant 4.84$.

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